

SYNOPSIS

Smart Village Dashboard

Submitted to

**LOVELY PROFESSIONAL UNIVERSITY PHAGWARA,
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TITLE

SMART VILLAGE DASHBOARD

1. Introduction / Background

Rural areas are of great importance to the development of a nation; however, villages are still struggling to make data-driven decisions and manage their resources effectively, as well as monitor their essential services such as agriculture, water supply, healthcare, etc., in a timely fashion. With the advent of Artificial Intelligence (AI) technologies and new web technologies, it is now possible to implement Smart Village systems that can effectively collect data, analyze data, and visualize data to help rural areas make better decisions for their development. The potential of MERN technologies implemented in Smart Village systems has shown promise to increase agricultural production, improve the usage of resources, and improve the quality of life of citizens through data-driven decisions.

The proposed project is a Smart Village Dashboard that uses web technologies such as React.js and Next.js for a responsive frontend, Node.js for backend APIs, MongoDB as a database, Redis for caching and session management, and JWT for authentication. The proposed project uses a combined AI layer that performs predictive analytics on the data of villages to make decisions on various domains such as agriculture, water supply and infrastructure planning, which aligns with the AI-driven rural initiatives.

The Smart Village Dashboard has been designed to be modular, scalable, and deployable within the Indian village context, with a special focus on usability for local administrators and stakeholders.

2. Literature Review

Current studies have proven that digital technologies and AI can be leveraged to accelerate rural development. Studies have been carried out to analyze the implementation of Big Data and AI technologies in smart villages. The studies have proven that it is possible to create quality life, avoid resource wastage, and create sustainable rural communities with the use of sensors and AI technologies.

Currently, web development has established effective technologies to create secure and high-performing dashboards. The technologies include the use of React.js and Next.js to create user interfaces, Node.js to execute server-side operations, JWT to implement authentication without session state, and Redis to implement session, cache, and security token management. Today, modern dashboard and analytics platforms have widely adopted these technologies to offer role-based views, real-time data updates, and intuitive user interactions.

However, current technologies have been developed to implement urban smart city applications, as well as focus on specific domains such as smart farming.

The majority of rural technology solutions rely on static reports, offer limited visualizations, or are unable to offer AI-driven forecasts and risks necessary for forward-thinking governance.

This project addresses this by utilizing AI technology and current full-stack web technology to develop an integrated Smart Village Dashboard that caters to rural decision-makers' needs by offering multiple user access and real-time data analysis capabilities.

3. Problem Statement

Local government agencies and leaders in rural areas tend to deal with incomplete, out-of-date, and fragmented information regarding village-level information on agricultural conditions, water resources, quality of infrastructure, public health standards, and execution of programs. This gives rise to a number of issues:

- Decisions are taken after problems arise, resulting in a lack of timely responses to new challenges.
- Challenges are encountered while trying to determine priorities for resource allocation.
- There is a lack of a centralized and user-friendly interface where all stakeholders can gain access to information as it becomes available.

Furthermore, with the increasing number of users accessing digital platforms from multiple locations and devices, there is an increasing need to develop strong authentication and access control systems that can effectively handle multiple types of users (such as system administrators, local officials, and residents) while providing optimal performance.

The fundamental question that this project aims to address is:

How can we develop an AI technology-based web-based Smart Village Dashboard using React.js, Next.js, Node.js, JWT, Redis, and data analytics technology to provide secure, real-time, and forecasting capabilities for rural decision-making processes?

4. Objectives

4.1 General Objective

To develop, design, and evaluate an "AI enabled Smart Village Dashboard" for data-driven governance and planning in the rural sector, utilizing a "Full Stack Web Development" approach.

4.2 Specific Objectives

1. To design a responsive, user-friendly frontend interface using React.js and Next.js frameworks for different user roles, i.e., administrators, village officers, and citizens.
 2. To design a Node.js backend system with secure REST APIs for data ingestion, processing, and retrieval from various sources, including IoT devices, government sources, and manual inputs.
 3. To design JWT-based authentication and Redis-based session/token management for secure user login, role-based access control, and performance benefits.
 4. To design a MongoDB-based data storage system to store data, including time series data, master data, user data, and AI results, for the village domain.
 5. To design a system to incorporate AI and data analytics models, including regression, anomaly detection, and prioritization, to produce predictive results for agriculture, water, and infrastructure planning.
 6. To evaluate the system in terms of usability, performance, security, and usefulness of the results obtained from the AI models for rural area decision makers.
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5. Research Methodology

5.1 Research Design

The initiative incorporates an applied research methodology in experimental software engineering. The project process is an iterative process with qualitative inputs from stakeholders (such as village officers and experts in the field) and quantitative evaluations of the performance of the systems and the accuracy of the AI model.

5.2 System Architecture and Technologies

The proposed architecture includes the following main layers:

- Data Sources:
 - Government and Open APIs for schemes, weather, statistics.
 - Manual data entry by village officers/administrators.
- Backend Layer (Node.js/Express):
 - API Gateway for data ingestion, querying, and management.
 - Business logic for validation, transformation, and aggregation of data from villages.
 - Integration of AI/ML engine for prediction.
- Authentication and Security:
 - JWT (JSON Web Tokens) is used for stateless authentication/authorization.
 - Redis is used as cache for storing frequently used data or token blacklisting/session management if required.
- Data Storage:
 - MongoDB is used as the database for storing structured/semi-structured data.
 - Sensor data, records, user data, AI/ML data.
- AI and Data Analytics Layer:
 - The AI/ML models are created to provide predictions on crop yields, water demand, risk of infrastructures, prioritization of development activities for villages.
 - Data preprocessing, feature engineering, model building/training/evaluation are done using ML tools.
 - The ML models are created as a service that can be called from the backend.
- Frontend Layer (React.js/Next.js):
 - Dashboards for various user roles.
 - Next.js is used for routing, server-side rendering, API routes.
 - Role-specific pages/views.
 - Dashboards for various user roles.
 - Role-specific

5.3 Development Phases

1. Requirement Analysis:

Functional and Non-Functional Requirements will be gathered through informal discussions and analysis of rural development use cases or existing Smart Village or Smart City Solutions.

2. System Design:

- Overall System Architecture, Database Schema, API Endpoints, and User Interface will be designed.
- Planning of JWT Role-Based Security, Redis Usage, and AI Integration.

3. Implementation:

- Implement Frontend using React.js and Next.js.
- Implement Backend using Node.js/Express.
- Connect Redis for Caching and Token/Session Features.
- Connect MongoDB as a Database.

4. AI/ML Integration:

- Collect Historical or Sample Datasets on Agriculture, Water Usage, or Infra-Related Metrics.
- Train AI Models (Regression, Classification, Anomaly Detection), etc., and Create a Service that the Backend Can Call to Generate Predictions.

5. Testing and Validation:

- Functional Testing of all Features.
- Performance Testing.
- Security Testing.

6. Evaluation:

- User Feedback from a Small Group of Test Users (e.g., Students, Mentors, Local Government Officials).
- System Metrics: Analysis of Overall System Metrics (e.g., Load Time, Error Rate), AI Metrics: Analysis of Overall AI Metrics (e.g., Accuracy, Precision, Recall).

7. Expected Outcomes

This project aims to achieve the following results:

- The Smart Village Dashboard will be implemented as a web application by the development team. This application will include features such as user authentication, role-based access control, and dynamic data visualization. The technology stack for the project will be React.js, Next.js, Node.js, JWT, Redis, and MongoDB.
 - The Smart Village Dashboard will be enriched with artificial intelligence features that can provide meaningful forecasts. This feature will be implemented for one or two critical areas, such as agriculture and water.
 - The project will be accompanied by complete documentation, which will describe the entire development process, the technology stack, and the evaluation criteria. This documentation will also show how the latest web development technologies and artificial intelligence can be used for decision-making processes.
 - The project will also include the following:
Future development possibilities will be identified, which can be useful for extending the Smart Village Dashboard for other villages and establishing connections with actual villages.
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